

9.3 State-space Establishing of Linear System

General Methodology:

1. From **Physics Mechanism** of System
2. From **Differential Equations** of System
3. From **Transfer Functions** of System
4. From **State-variable Diagram** of System
5. **Linear Transformation of State space**

9.3.1 From Physics Mechanism of System

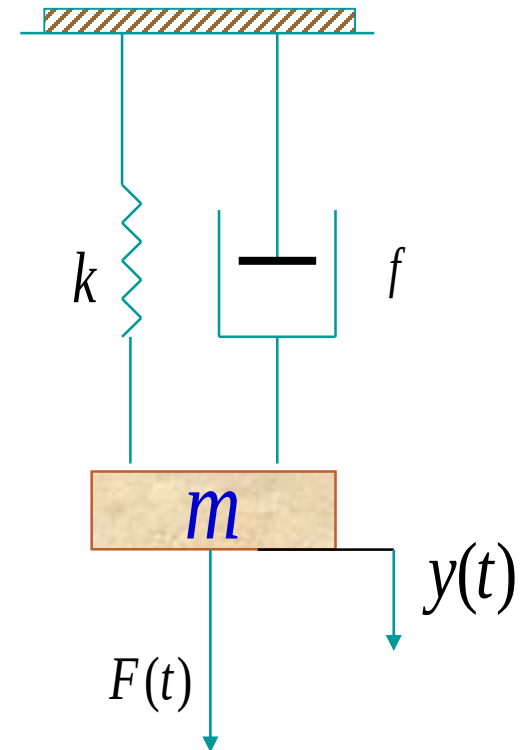
Ex.9-2 Mechanism system composed by mass, spring(彈簧) and damper(阻尼器) without gravity(重力).

from Newton's law

$$m \frac{d^2 y}{dt^2} + f \frac{dy}{dt} + ky = F(t)$$

in which, $F(t)$ is Input, $y(t)$ is Output.

Then, if the original displacement and velocity are available, the system's solution of the certain input is available as well.



Select the **displacement** and **velocity** as the **state variables**

$$x_1 = y, \quad x_2 = v = \dot{y}$$

Input is:

$$u(t) = F(t)$$

State-equations:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{k}{m}x_1 - \frac{f}{m}x_2 + \frac{1}{m}u \end{cases}$$

State space representation:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{f}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Ex.9-3 The **State-space representation** of Mechanism without gravity, the **input** is the pull F , the **outputs** are the displacement y_1 and y_2 .

From Newton's first Law, we have physics relationship of m_1 and m_2 :

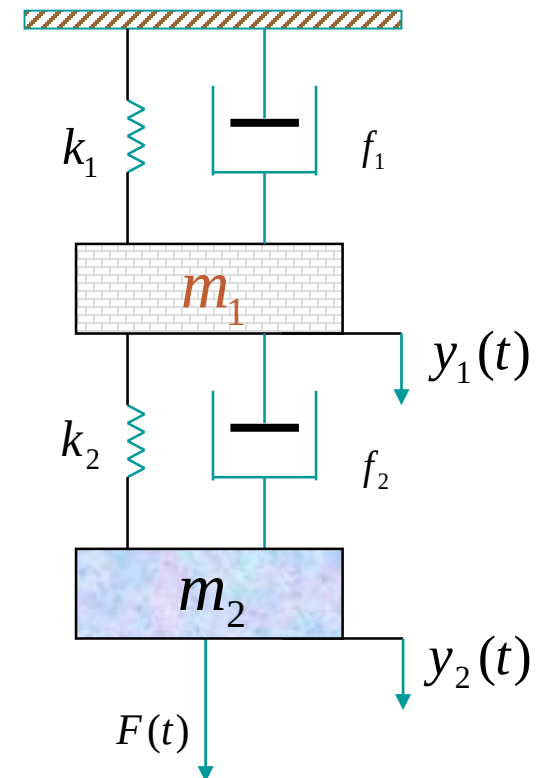
$$m_2 \ddot{y}_2 = F(t) - k_2(y_2 - y_1) - f_2(\dot{y}_2 - \dot{y}_1)$$

$$m_1 \ddot{y}_1 = k_2(y_2 - y_1) + f_2(\dot{y}_2 - \dot{y}_1) - k_1 y_1 - f_1 \dot{y}_1$$

select 4 independent state variables:

$$x_1 = y_1, \quad x_2 = y_2, \quad x_3 = \dot{y}_1, \quad x_4 = \dot{y}_2$$

$$\begin{cases} \dot{x}_1 = x_3 \\ \dot{x}_2 = x_4 \\ \dot{x}_3 = \frac{k_2}{m_1}(x_2 - x_1) + \frac{f_2}{m_1}(x_4 - x_3) - \frac{k_1}{m_1}x_1 - \frac{f_1}{m_1}x_3 \\ \dot{x}_4 = \frac{1}{m_2}F(t) - \frac{k_2}{m_2}(x_2 - x_1) - \frac{f_2}{m_2}(x_4 - x_3) \end{cases}$$



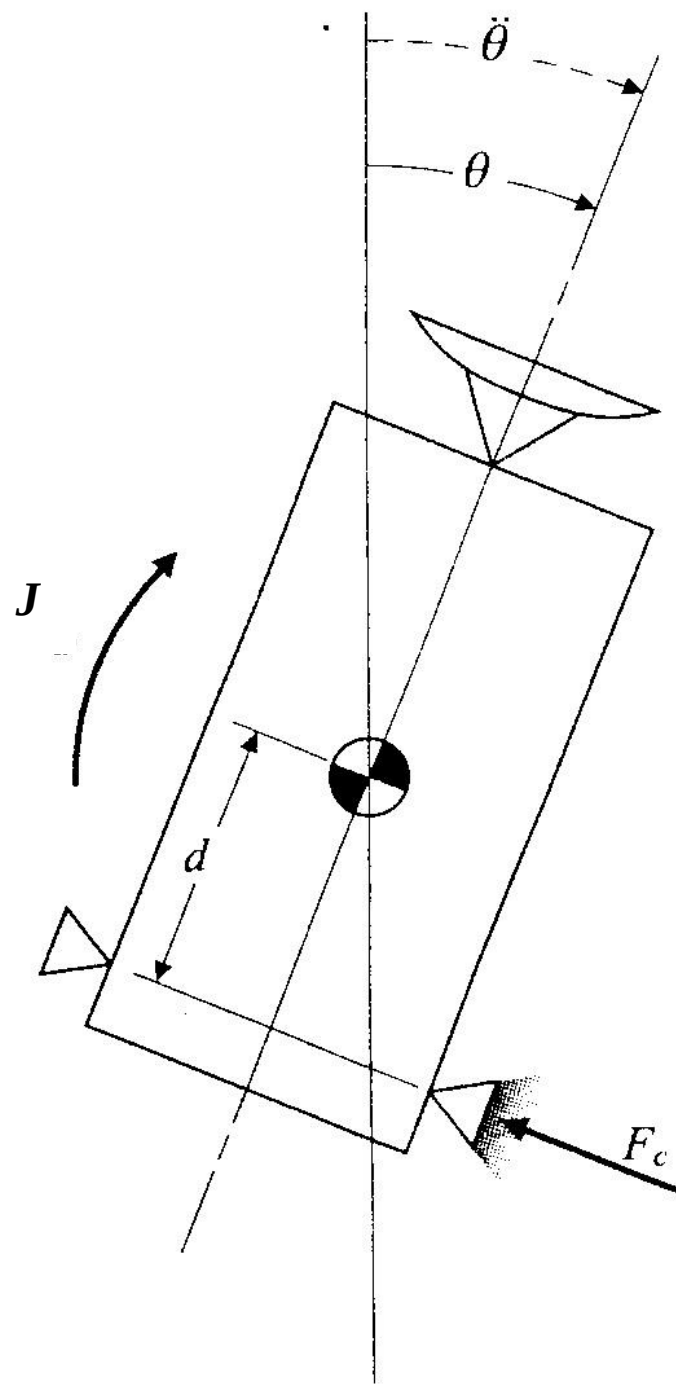
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{k_1 + k_2}{m_1} & \frac{k_2}{m_1} & -\frac{f_1 + f_2}{m_1} & \frac{f_2}{m_1} \\ \frac{k_2}{m_2} & -\frac{k_2}{m_2} & \frac{f_2}{m_2} & -\frac{f_2}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_2} \end{bmatrix} F$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

A

B

C



9.3 State-space Establishing of Linear System

General Methodology:

1. From **Physics Mechanism** of System
2. From **Differential Equations** of System
3. From **Transfer Functions** of System
4. From **State-variable Diagram** of System
5. **Linear Transformation of State space**

9.3.2 From Differential Equations of System

➤ Methodology:

- Establish the differential/difference equations by the physics mechanism of system;
- Establish the state equation focusing on the equations and a group of state-variables;
- Establish the output function based on the relationship between system's outputs and states.

➤ State-variable Selection

- ✓ Selection of state variable is not unique.
- ✓ Methodology:
 - ☞ 1. Select variable in the **initial conditions or related**.
 - ☞ 2. Select characteristic variable of independent storage components (energy or information) **with certain physical meaning**, such as the electric current i of inductance, the voltage u_c of capacitor, and velocity v of mass, etc.

Scenario (1): No derivatives(微分) of input u contained in n-order linear differential equations

Assume the dynamic process of the SISO control system is described as follow:

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} \dot{y} + a_n y = bu$$

$y^{(n)}, y^{(n-1)}, \cdots, \dot{y}, y$ ——— Derivatives of output

u ——— Input

If initial conditions $y(0), y'(0), \dots, y^{(n-1)}(0)$ of output and the input $u(t)$ of $t \geq 0$ are known, the behavior of system at any time can be obtained.

Select state variables:

$$\begin{cases} x_1 = y \\ x_2 = \dot{y} \\ \vdots \\ x_{n-1} = y^{(n-2)} \\ x_n = y^{(n-1)} \end{cases}$$

then:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \vdots \\ \dot{x}_{n-1} = x_n \\ \dot{x}_n = -a_n x_1 - a_{n-1} x_2 - \cdots - a_1 x_n + bu \end{cases}$$

therefore:

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} \dot{y} + a_n y = bu$$

State space:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x}$$

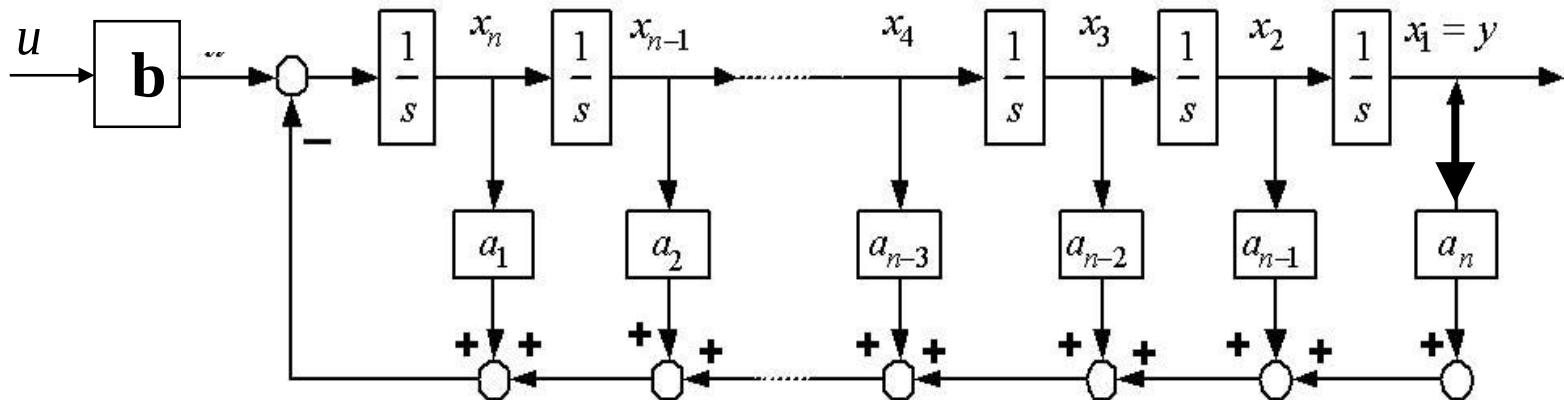
in which, the matrices:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ b \end{bmatrix}$$

$$\mathbf{C} = [1 \ 0 \ 0 \ \cdots \ 0]$$

➤ Then, draw the following **block diagram** (**state variable diagram**) among the state variables.

- ✓ The output of each integrator corresponds to each **state variable**.
- ✓ The **state equations** are decided by the relationship of I/O.
- ✓ The **output equation** is on output part.



$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = bu$$

Ex.9-4 assume the differential equation of the system's dynamic process is

$$\ddot{y} + 6\ddot{y} + 11\dot{y} + 6y = 6u$$

in which, u and y are input and output.

Try to find the state space representation of the system.

Select state-variables: $x_1 = y, x_2 = \dot{y}, x_3 = \ddot{y},$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \dot{x}_3 = -6x_1 - 11x_2 - 6x_3 + 6u \end{cases}$$

Standard Form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} u$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x}$$

A

B

C

$$y = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Scenario(2): Derivatives of input u contained in n-order linear differential equation system

n-order linear differential equation representation:

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} \dot{y} + a_n y = b_o u^{(n)} + b_1 u^{(n-1)} + \cdots + b_{n-1} \dot{u} + b_n u$$

Reference scenario(1):

$$\left\{ \begin{array}{l} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \vdots \\ \dot{x}_{n-1} = x_n \\ \dot{x}_n = -a_n y - a_{n-1} \dot{y} - \cdots - a_1 y^{(n-1)} + b_o u^{(n)} + b_1 u^{(n-1)} + \cdots + b_{n-1} \dot{u} + b_n u \\ \quad = -a_n x_1 - a_{n-1} x_2 - \cdots - a_1 x_n + b_o u^{(n)} + b_1 u^{(n-1)} + \cdots + b_{n-1} \dot{u} + b_n u \end{array} \right. \quad \leftarrow \quad \left\{ \begin{array}{l} x_1 = y \\ x_2 = \dot{y} \\ \vdots \\ x_{n-1} = y^{(n-2)} \\ x_n = y^{(n-1)} \end{array} \right.$$

However, the derivative of input u is still contained in state equation, which is **INCONSEQUENCE(不合理)**.



Analysis: if input u is a limitary Step signal: Step Function, u' will be the Impulsive Function δ , $u^{(i)}(i=2,3,\dots)$ will be the higher-order impulse function, and state trajectory will have infinite jump at t_0 .

Hence, we cannot choose the output y and its derivatives to be state variables of the system. Such group of state variables cannot decide the future state of the system based on the known system input and original state conditions.

The principle of state variable selection:

No derivative of the input/operation function could be included in any differential function in the system state equations represented by first-order differential equation sets.

Select the state variables:

The input is contained in the state variables

n-1 state equations

$$\left\{ \begin{array}{l} x_1 = y - \beta_0 u \\ x_2 = \dot{y} - \beta_0 \dot{u} - \beta_1 u = \dot{x}_1 - \beta_1 u \\ x_3 = \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 u = \dot{x}_2 - \beta_2 u \\ \vdots \\ x_n = y^{(n-1)} - \beta_0 u^{(n-1)} - \beta_1 u^{(n-2)} - \dots - \beta_{n-2} \dot{u} - \beta_{n-1} u = \dot{x}_{n-1} - \beta_{n-1} u \end{array} \right.$$

then

$$\left\{ \begin{array}{l} \dot{x}_1 = x_2 + \beta_1 u \\ \dot{x}_2 = x_3 + \beta_2 u \\ \vdots \\ \dot{x}_{n-1} = x_n + \beta_{n-1} u \\ \dot{x}_n = ? \quad ? \quad ? \end{array} \right.$$

How to find the relationship between $\dot{x}_n$ and other states: $x_1, x_2,$

$\dots, x_{n-1}$

? ?

Solution: $x'_n = f(x_1, x_2, \dots, x_{n-1}, u)$

state equation of x_n : $x_n = y^{(n-1)} - \beta_0 u^{(n-1)} - \beta_1 u^{(n-2)} - \dots - \beta_{n-2} \dot{u} - \beta_{n-1} u$

derivative of x_n : $\dot{x}_n = y^{(n)} - \beta_0 u^{(n)} - \beta_1 u^{(n-1)} - \dots - \beta_{n-2} \ddot{u} - \beta_{n-1} \dot{u}$

differential equation: $y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} \dot{u} + b_n u$

and then: $y^{(n)} = -a_1 y^{(n-1)} - \dots - a_{n-1} \dot{y} - a_n y + b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} \dot{u} + b_n u$

substitute $y^{(n)}$ into x'_n :

$$\begin{aligned}
 \dot{x}_n &= y^{(n)} - \beta_0 u^{(n)} - \beta_1 u^{(n-1)} - \dots - \beta_{n-1} \dot{u} \\
 &= (-a_1 y^{(n-1)} - a_2 y^{(n-2)} - \dots - a_{n-1} \dot{y} - a_n y + b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} \dot{u} + b_n u) \\
 &\quad - \beta_0 u^{(n)} - \beta_1 u^{(n-1)} - \dots - \beta_{n-1} \dot{u} \\
 &= -a_1 y^{(n-1)} - a_2 y^{(n-2)} - \dots - a_{n-1} \dot{y} - a_n y \\
 &\quad + (b_0 - \beta_0) u^{(n)} + (b_1 - \beta_1) u^{(n-1)} + \dots + (b_{n-1} - \beta_{n-1}) \dot{u} + b_n u
 \end{aligned}$$

state variables:
$$\begin{cases} x_1 = y - \beta_0 u \\ x_2 = \dot{y} - \beta_0 \dot{u} - \beta_1 u \\ x_3 = \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 u \\ \vdots \\ x_n = y^{(n-1)} - \beta_0 u^{(n-1)} - \beta_1 u^{(n-2)} - \dots - \beta_{n-2} \dot{u} - \beta_{n-1} u \end{cases}$$

derivatives of y:
$$\begin{cases} y = x_1 + \beta_0 u \\ \dot{y} = x_2 + \beta_0 \dot{u} + \beta_1 u \\ \ddot{y} = x_3 + \beta_0 \ddot{u} + \beta_1 \dot{u} + \beta_2 u \\ \vdots \\ y^{(n-1)} = x_n + \beta_0 u^{(n-1)} + \beta_1 u^{(n-2)} + \dots + \beta_{n-2} \dot{u} + \beta_{n-1} u \end{cases}$$

substitute output y and its derivatives: $y', \dots, y^{(n-1)}$ into x'_n :

$$\begin{aligned}\dot{x}_n = & -a_n x_1 - a_{n-1} x_2 - \cdots - a_2 x_{n-1} - a_1 x_n \\ & + (b_0 - \beta_0) u^{(n)} + (b_1 - \beta_1 - a_1 \beta_0) u^{(n-1)} + (b_2 - \beta_2 - a_1 \beta_1 - a_2 \beta_0) u^{(n-2)} + \cdots \\ & + (b_n - a_1 \beta_{n-1} - a_2 \beta_{n-2} - \cdots - a_{n-1} \beta_1 - a_n \beta_0) u\end{aligned}$$

Principle:

No derivative of input $u(t)$ contained in state equations.

thus:

$$\begin{cases} \beta_0 = b_0 \\ \beta_1 = b_1 - a_1 \beta_0 \\ \beta_2 = b_2 - a_1 \beta_1 - a_2 \beta_0 \\ \vdots \\ \beta_n = b_n - a_1 \beta_{n-1} - \cdots - a_{n-1} \beta_1 - a_n \beta_0 \end{cases}$$

State-space of the system is:

$$\begin{cases} \dot{x}_1 = x_2 + \beta_1 u \\ \dot{x}_2 = x_3 + \beta_2 u \\ \vdots \\ \dot{x}_{n-1} = x_n + \beta_{n-1} u \\ \dot{x}_n = -a_n x_1 - a_{n-1} x_2 - \cdots - a_1 x_n + \beta_n u \end{cases}$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

Rewrite the system to the matrix representation:

$$y = \mathbf{C}\mathbf{x} + \mathbf{D}u$$

In which:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}$$

$$\mathbf{C} = [1 \quad 0 \quad 0 \quad \cdots \quad 0]$$

$$\mathbf{D} = \beta_0 = b_0$$

Ex.9-5 Assume the dynamic equation of a control system can be written as the differential equation:

$$\ddot{y} + 6\dot{y} + 11y = 11\dot{u} + 6u$$

try to give its state space description.

Solution: compare with the standard differential equation

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \dots + b_{n-1} \dot{u} + b_n u$$

we have:

$$a_1 = 6, \quad a_2 = 11, \quad a_3 = 2,$$

$$b_0 = 0, \quad b_1 = 0, \quad b_2 = 11, \quad b_3 = 6$$

$$x_1 = y - \beta_0 u$$

$$x_2 = \dot{x}_1 - \beta_1 u$$

$$x_3 = \dot{x}_2 - \beta_2 u$$

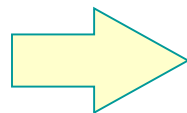
and the coefficients:

$$\beta_0 = b_0 = 0$$

$$\beta_1 = b_1 - a_1 \beta_0 = 0$$

$$\beta_2 = b_2 - a_1 \beta_1 - a_2 \beta_0 = 11$$

$$\beta_3 = b_3 - a_1 \beta_2 - a_2 \beta_1 - a_3 \beta_0 = -60$$



$$x_1 = y$$

$$x_2 = \dot{x}_1$$

$$x_3 = \dot{x}_2 - 11u$$

State equations are:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3 + 11u$$

$$\dot{x}_3 = -a_3x_1 - a_2x_2 - a_1x_3 + \beta_3u = -2x_1 - 11x_2 - 6x_3 - 60u$$

State space description of matrix:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 11 \\ -60 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The standard form:

$$y^{(n)}(t) + a_1 y^{(n-1)}(t) + \cdots + a_{n-1} \dot{y}(t) + a_n y(t) = b_0 u^{(n)} + b_1 u^{(n-1)} + \cdots + b_{n-1} u' + b_n u(t)$$

while if $b_0=0$, we can select another group of state variable as follow:

$$x_n = y$$

$$x_i = \dot{x}_{i+1} + a_{n-i} y - b_{n-i} u \quad i = 1, 2, 3, \dots, n-1$$



$$x_{n-1} = \dot{x}_n + a_1 y - b_1 u$$

$$\dot{x}_n = x_{n-1} - a_1 x_n + b_1 u$$

$$x_{n-2} = \dot{x}_{n-1} + a_2 y - b_2 u$$

$$\dot{x}_{n-1} = x_{n-2} - a_2 x_n + b_2 u$$

$\vdots$



$\vdots$

$\dot{x}_1 ???$

$$x_2 = \dot{x}_3 + a_{n-2} y - b_{n-2} u$$

$$\dot{x}_3 = x_2 - a_{n-2} x_n + b_{n-2} u$$

$$x_1 = \dot{x}_2 + a_{n-1} y - b_{n-1} u$$

$$\dot{x}_2 = x_1 - a_{n-1} x_n + b_{n-1} u$$

Output equation: $y = x_n$

Furthermore:

$$x_{n-1} = \dot{x}_n + a_1 y - b_1 u$$

$$= \dot{y} + a_1 y - b_1 u$$

$$x_{n-2} = \dot{x}_{n-1} + a_2 y - b_2 u$$

$$= \ddot{y} + a_1 \dot{y} - b_1 \dot{u} + a_2 y - b_2 u$$

$\vdots$

$$x_2 = \dot{x}_3 + a_{n-2} y - b_{n-2} u$$

$$= y^{(n-2)} + a_1 y^{(n-3)} - b_1 u^{(n-3)} + a_2 y^{(n-4)} - b_2 u^{(n-4)} + \dots + a_{n-2} y - b_{n-2} u$$

$$x_1 = \dot{x}_2 + a_{n-1} y - b_{n-1} u$$

$$= y^{(n-1)} + a_1 y^{(n-2)} - b_1 u^{(n-2)} + a_2 y^{(n-3)} - b_2 u^{(n-3)} + \dots + a_{n-1} y - b_{n-1} u$$

$$\begin{aligned}
 x_1 &= \dot{x}_2 + a_{n-1}y - b_{n-1}u \\
 &= y^{(n-1)} + a_1y^{(n-2)} - b_1u^{(n-2)} + a_2y^{(n-3)} - b_2u^{(n-3)} + \dots + a_{n-1}y - b_{n-1}u
 \end{aligned}$$

calculate derivate of x_1



$$\dot{x}_1 = y^{(n)} + a_1y^{(n-1)} - b_1u^{(n-1)} + a_2y^{(n-2)} - b_2u^{(n-2)} + \dots + a_{n-1}\dot{y} - b_{n-1}\dot{u}$$

bring $y^{(n)}$ into x'_1 according to:

$$y^{(n)}(t) + a_1y^{(n-1)}(t) + \dots + a_{n-1}\dot{y}(t) + a_ny(t) = b_0u^{(n)} + b_1u^{(n-1)} + \dots + b_{n-1}\dot{u} + b_nu$$



$$\dot{x}_1 = -a_n x_n + b_n u$$

Matrix description:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}u$$

$$y = \mathbf{C}\mathbf{x} + \mathbf{d}u$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ 0 & 1 & \cdots & 0 & -a_{n-2} \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 1 & -a_1 \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} b_n \\ b_{n-1} \\ \vdots \\ b_1 \end{bmatrix}$$

$$\mathbf{C} = [0 \quad 0 \quad \cdots \quad 1]$$

$$\mathbf{d} = 0$$

Ex.9-5(II) Differential equation of control system is

$$\ddot{y} + 6\ddot{y} + 11\dot{y} + 2y = 11\dot{u} + 6u$$

try to give its state space description.

Solution: standard differential equation

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} \dot{y} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \cdots + b_{n-1} \dot{u} + b_n u$$

$$a_1 = 6, \quad a_2 = 11, \quad a_3 = 2,$$

$$b_0 = 0, \quad b_1 = 0, \quad b_2 = 11, \quad b_3 = 6$$

select state variable

$$x_3 = y$$

$$x_2 = \dot{x}_3 + a_1 y - b_1 u$$

$$x_1 = \dot{x}_2 + a_2 y - b_2 u$$



$$x_3 = y$$

$$x_2 = \dot{x}_3 - 6y$$

$$x_1 = \dot{x}_2 - 11y + 11u$$

State space description:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 0 & -11 \\ 0 & 1 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Conclusion: for a certain system, selection of state variables is not unique.

Ex.9-6 The equations of a 2input/2output 2nd-order system are:

$$\ddot{y}_1 + a_1 \dot{y}_1 + a_2 y_2 = b_1 \dot{u}_1 + b_2 u_1 + b_3 u_2$$

$$\dot{y}_2 + a_3 y_2 + a_4 y_1 = b_4 u_2$$

try to give its state space description.

Solution: find derivative of y_1 , y_2 with highest-order

$$\ddot{y}_1 = -a_1 \dot{y}_1 + b_1 \dot{u}_1 - a_2 y_2 + b_2 u_1 + b_3 u_2$$

$$\dot{y}_2 = -a_3 y_2 - a_4 y_1 + b_4 u_2$$

calculate their integration

$$\begin{aligned} y_1 &= \iint [(-a_1 \dot{y}_1 + b_1 \dot{u}_1) + (-a_2 y_2 + b_2 u_1 + b_3 u_2)] dt^2 \\ &= \int (-a_1 y_1 + b_1 u_1) dt + \iint (-a_2 y_2 + b_2 u_1 + b_3 u_2) dt^2 \\ &= \int [(-a_1 y_1 + b_1 u_1) + \int (-a_2 y_2 + b_2 u_1 + b_3 u_2) dt] dt \\ y_2 &= \int [(-a_3 y_2 - a_4 y_1 + b_4 u_2)] dt \end{aligned}$$

Select state variables: $x_1 = y_1$

$$x_2 = y_2$$

From the equation of y_1 :

$$\begin{aligned}\dot{x}_1 &= -a_1 y_1 + b_1 u_1 + \int (-a_2 y_2 + b_2 u_1 + b_3 u_2) dt \\ &= -a_1 x_1 + b_1 u_1 + \int (-a_2 x_2 + b_2 u_1 + b_3 u_2) dt\end{aligned}$$

Select another state variable:

$$x_3 = \int (-a_2 x_2 + b_2 u_1 + b_3 u_2) dt$$

and

$$\dot{x}_3 = -a_2 x_2 + b_2 u_1 + b_3 u_2$$

From the equation of y_2 :

$$\dot{x}_2 = -a_3 x_2 - a_4 x_1 + b_4 u_2$$

The equation set: $\dot{x}_1 = -a x_1 + x_3 + b_1 u_1$

$$\dot{x}_2 = -a_4 x_1 - a_3 x_2 + b_4 u_2$$

$$\dot{x}_3 = -a_2 x_2 + b_2 u_1 + b_3 u_2$$

$$\dot{x}_1 = -a_1 x_1 + x_3 + b_1 u_1$$

$$\dot{x}_2 = -a_4 x_1 - a_3 x_2 + b_4 u_2$$

$$\dot{x}_3 = -a_2 x_2 + b_2 u_1 + b_3 u_2$$

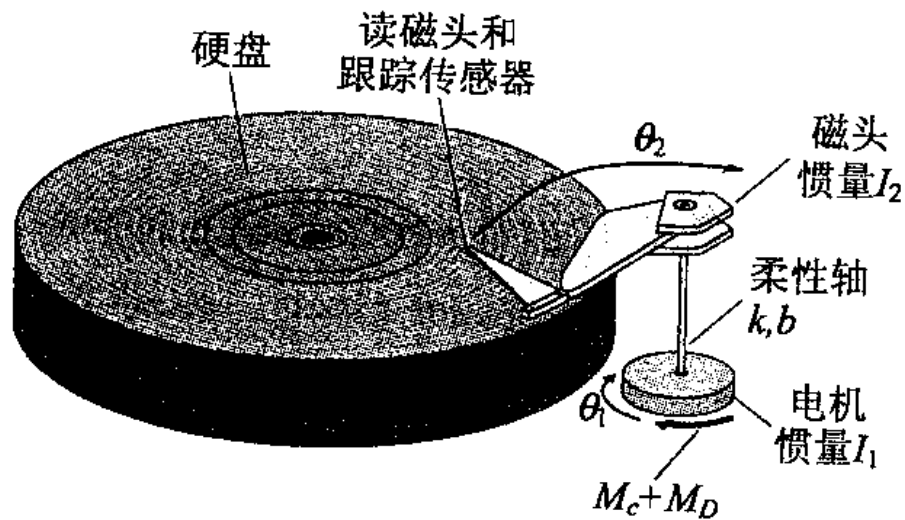
Rewrite the equations by the matrixes:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -a_1 & 0 & 1 \\ -a_4 & -a_3 & 0 \\ 0 & -a_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} b_1 & 0 \\ 0 & b_4 \\ b_2 & b_3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

The output matrix equation:

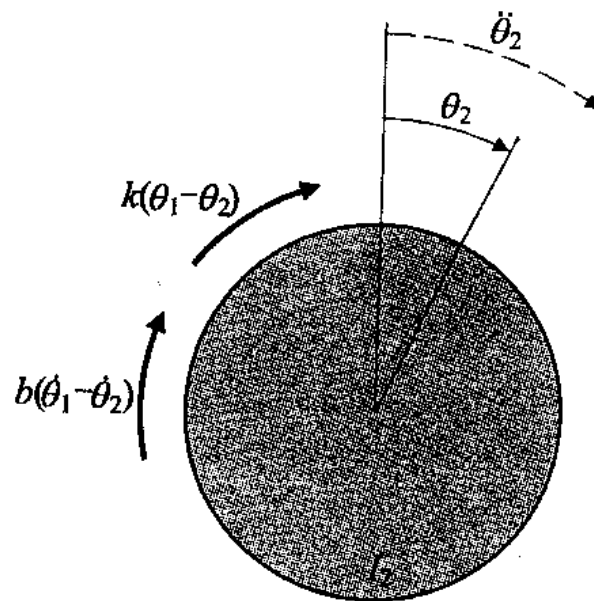
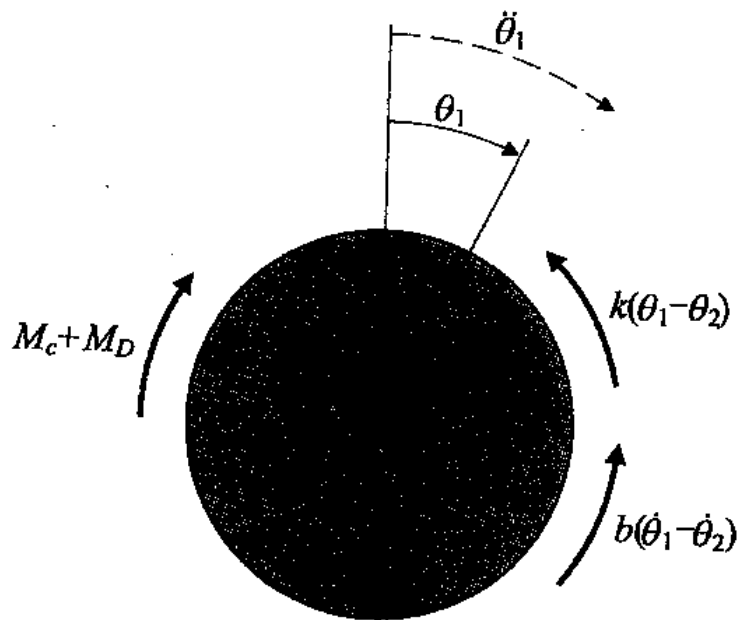
$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$





$$J_1 \ddot{\theta}_1 = M_c + M_D - k(\theta_1 - \theta_2) - b(\dot{\theta}_1 - \dot{\theta}_2)$$

$$J_2 \ddot{\theta}_2 = k(\theta_1 - \theta_2) + b(\dot{\theta}_1 - \dot{\theta}_2)$$



9.3 State-space Establishing of Linear System

General Methodology:

1. From **Physics Mechanism** of System
2. From **Differential Equations** of System
3. From **Transfer Functions** of System
4. From **State-variable Diagram** of System
5. **Linear Transformation of State space**

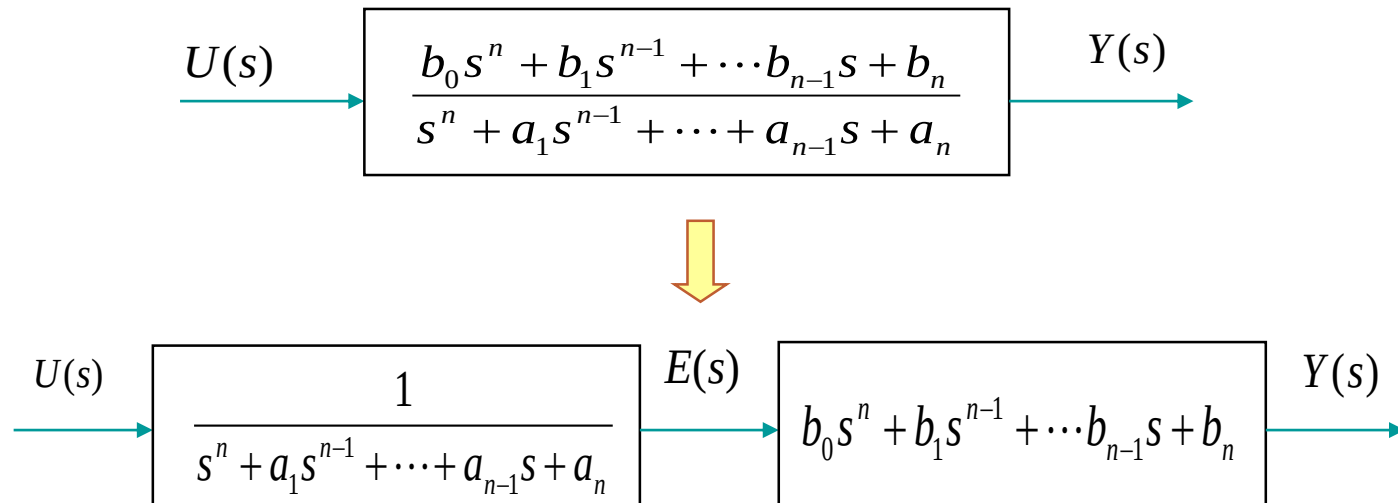
9.3.3 From Transfer Functions of System

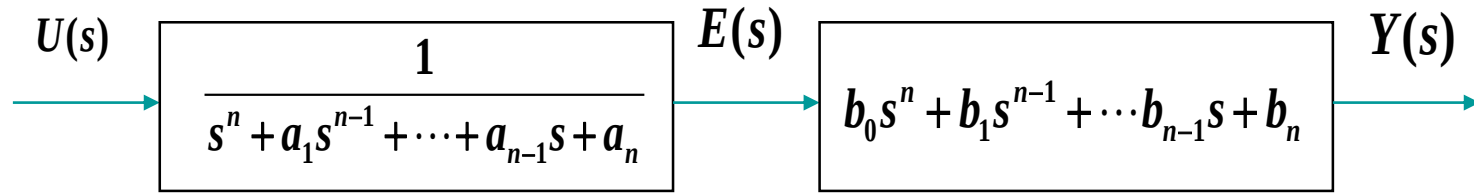
- Transfer Functions
- State Space



1. Transfer Functions to State Space

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$





$$U(s) = (s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n) E(s)$$

$$Y(s) = (b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n) E(s)$$

Select state variables:

$$\begin{cases} x_1 = e(t) \\ x_2 = \dot{e}(t) \\ \vdots \\ x_n = e^{(n-1)}(t) \end{cases} \quad \longrightarrow \quad \begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 \\ \vdots \\ \dot{x}_{n-1} = x_n \end{cases}$$

$$u = \dot{x}_n + a_1 x_n + a_2 x_{n-1} + \dots + a_{n-1} x_2 + a_n x_1$$

$$y = b_0 \dot{x}_n + b_1 x_n + b_2 x_{n-1} + \dots + b_{n-1} x_2 + b_n x_1$$

$$u = \dot{x}_n + a_1 x_n + a_2 x_{n-1} + \cdots + a_{n-1} x_2 + a_n x_1$$

$$y = b_0 \dot{x}_n + b_1 x_n + b_2 x_{n-1} + \cdots + b_{n-1} x_2 + b_n x_1$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} u$$

B

A

$$y = b_0 (-a_n \quad -a_{n-1} \quad \cdots \quad -a_1) \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u + (b_n \quad b_{n-1} \quad \cdots \quad b_1) \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

C

D

If $b_0=0$, the output equation will be simplified.

For the situation that the derivative of input u is included in the differential equations.

Homework: Comparing with P24

Ex.9-7 Transfer function of a control system is:

$$\frac{Y(s)}{U(s)} = \frac{s^2 + 4s + 1}{s^3 + 9s^2 + 8s}$$

transform it to the state space representation.

Solution:

$$a_1 = 9, \quad a_2 = 8, \quad a_3 = 0, \quad b_0 = 0, \quad b_1 = 1, \quad b_2 = 4, \quad b_3 = 1$$

The state equation is:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -8 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

The output equations is:

$$y = \begin{bmatrix} 1 & 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

➤ **Serialization of the Transfer Function:**

$$G(s) = \frac{Num(s)}{Den(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0} \quad m \leq n$$

The Numerator is: $Num = b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0$

The Denominator is: $Den = s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0$

If $z_1, z_2, \dots, z_m$ are m zero-points of $G(s)$,

and $p_1, p_2, \dots, p_n$ are n pole-points of $G(s)$.

then $G(s)$ is:

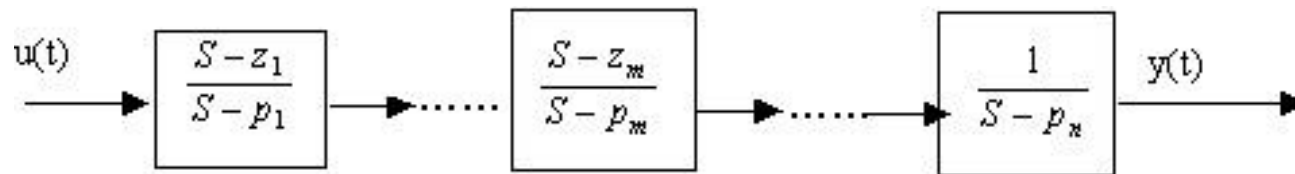
$$G(s) = \frac{b_m (s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_n)}$$

$$= \frac{s - z_1}{s - p_1} \bullet \frac{s - z_2}{s - p_2} \bullet \dots \bullet \frac{s - z_m}{s - p_m} \bullet \frac{b_m}{s - p_{m+1}} \bullet \dots \bullet \frac{1}{s - p_n}$$

Therefore the system is composed **serially** by n items below:

$$\frac{s - z_1}{s - p_1}, \frac{s - z_i}{s - p_i}, \dots, \frac{1}{s - p_n}$$

The structure of the system can be described by figure (a)。

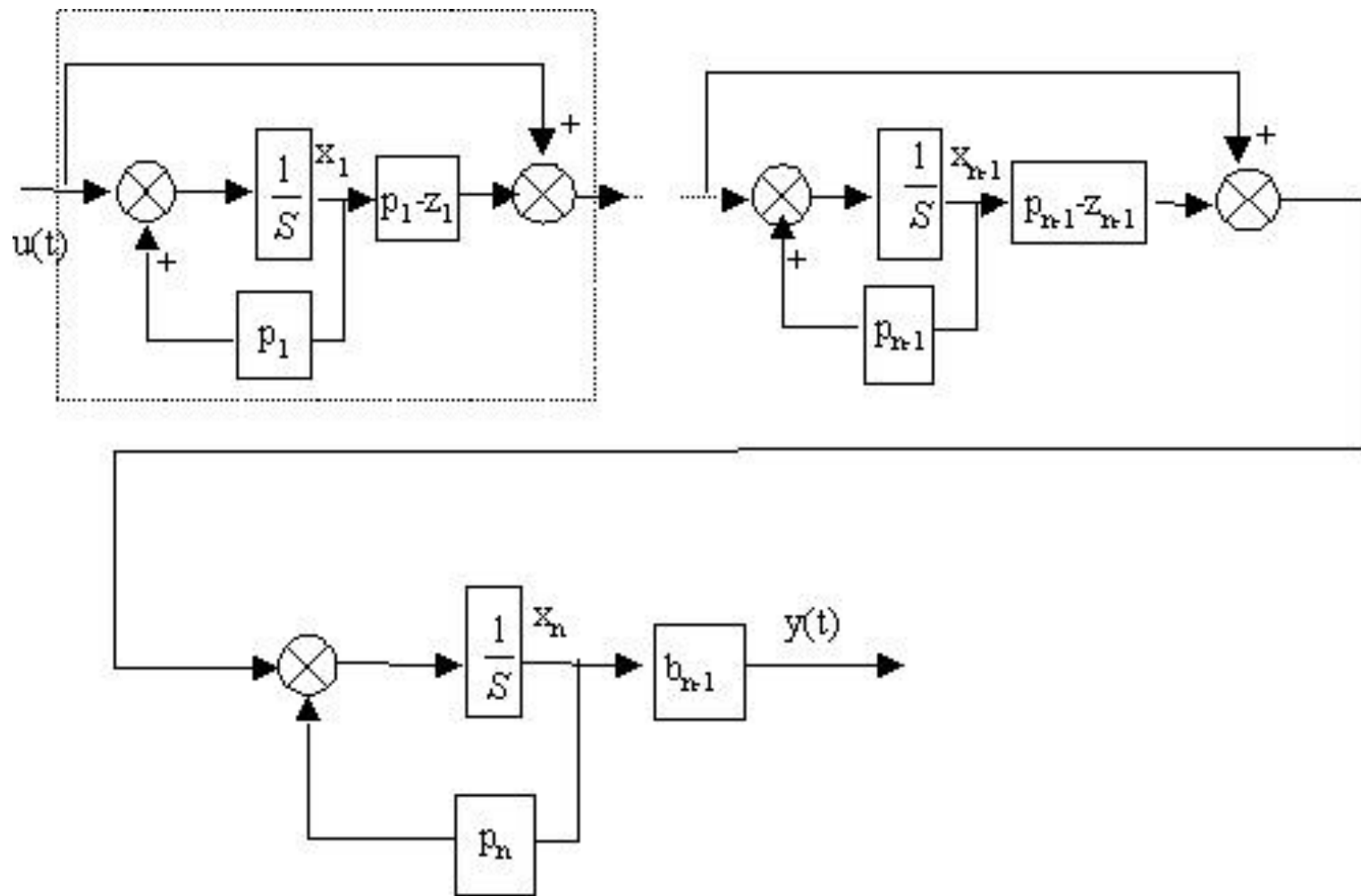


(a)

of which the first block is transformed as follow:

$$\frac{s - z_1}{s - p_1} = 1 + \frac{p_1 - z_1}{s - p_1} = 1 + (p_1 - z_1) \bullet \frac{1}{s} \frac{1}{1 - p_1 \frac{1}{s}}$$

Thus its structure block will be recomposed as in figure (b).



(b) ($m=n-1$)

Assume the outputs of integral items are required state variables
The state equations of the system are:

$$\begin{cases} \dot{x}_1 = p_1 x_1 + u \\ \dot{x}_2 = (p_1 - z_1)x_1 + u + p_2 x_2 = (p_1 - z_1)x_1 + p_2 x_2 + u \\ \vdots \\ \dot{x}_n = (p_1 - z_1)x_1 + (p_2 - z_2)x_2 + \cdots + (p_{n-1} - z_{n-1})x_{n-1} + p_n x_n + u \\ y = b_m x_n = b_{n-1} x_n, (m = n-1) \end{cases}$$

And the matrix representation:

$$\begin{cases} \dot{X} = \begin{bmatrix} p_1 & 0 & 0 & \cdots & 0 \\ p_1 - z_1 & p_2 & 0 & \cdots & 0 \\ p_1 - z_1 & p_2 - z_2 & p_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ p_1 - z_1 & p_2 - z_2 & p_3 - z_3 & \cdots & p_n \end{bmatrix} X + \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u \\ y = [0 \ 0 \ \cdots \ 0 \ b_{n-1}] X \end{cases}$$

➤ Parallel of the Transfer Function:

$$G(s) = \frac{Num(s)}{Den(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0} \quad m \leq n$$

The Denominator:

$$Den(s) = s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0$$

is the Characteristic Equations of the system.

Assume there are n characteristics roots: $p_i, i = 1, 2, \dots, n$

G(s) can be decomposed by the summation of n fractions:

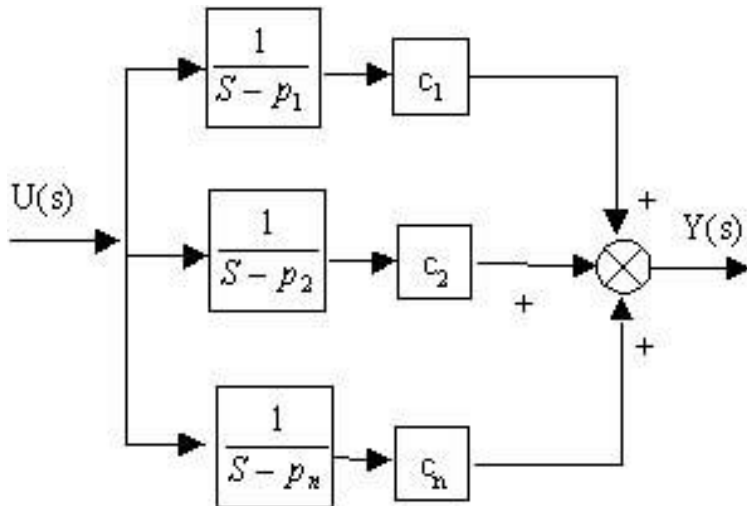
$$\begin{aligned} G(s) &= \frac{Y(s)}{U(s)} = \frac{c_1}{s - p_1} + \frac{c_2}{s - p_2} + \dots + \frac{c_n}{s - p_n} \\ &= \sum_{i=1}^n \frac{c_i}{s - p_i} \end{aligned}$$

In which, $c_i = \lim_{s \rightarrow p_i} (s - p_i) G(s)$, is called Residue (留数) of pole p_i .

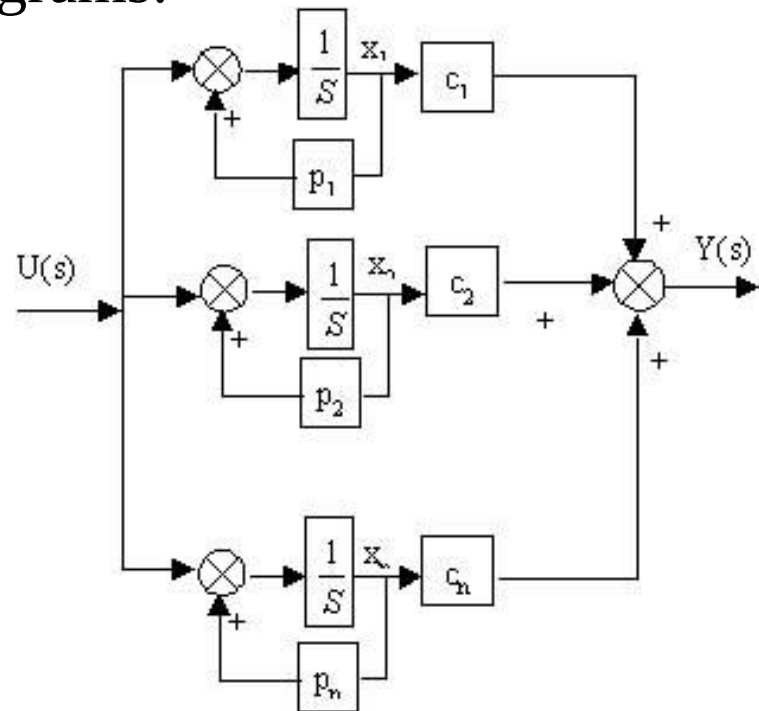
$$Y(s) = \frac{c_1}{s - p_1} U(s) + \frac{c_2}{s - p_2} U(s) + \dots + \frac{c_n}{s - p_n} U(s)$$

$$= \sum_{i=1}^n \frac{c_i}{s - p_i} U(s)$$

The parallel connection diagrams:



(a)



(b) Parallel connection
(No repeated roots)

The state-equations of figure (b):

The output equation:

$$y = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$$

$$\begin{cases} \dot{x}_1 = p_1 x_1 + u \\ \dot{x}_2 = p_2 x_2 + u \\ \vdots \\ \dot{x}_n = p_n x_n + u \end{cases}$$

Matrix representation:

$$\begin{cases} \dot{X} = \begin{bmatrix} p_1 & 0 & \dots & 0 \\ 0 & p_2 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & p_n \end{bmatrix} X + \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u \\ Y = [c_1 \ c_2 \ \dots \ c_n] X \end{cases}$$

PS: the system matrix A is a diagonal matrix.

If the denominator of $G(s)$ $Den(s)=0$ has repeated roots:

$$Den(s) = (s - p_1)^q (s - p_{q+1}) \cdots (s - p_n)$$

$s = p_1$ is the only q times repeated root, we have $G(s)$:

$$G(s) = \frac{Num(s)}{Den(s)}$$

$$= \frac{c_{11}}{s - p_1} + \frac{c_{12}}{(s - p_1)^2} + \dots + \frac{c_{1q}}{(s - p_1)^q} + \frac{c_{q+1}}{s - p_{q+1}} + \dots + \frac{c_n}{s - p_n}$$

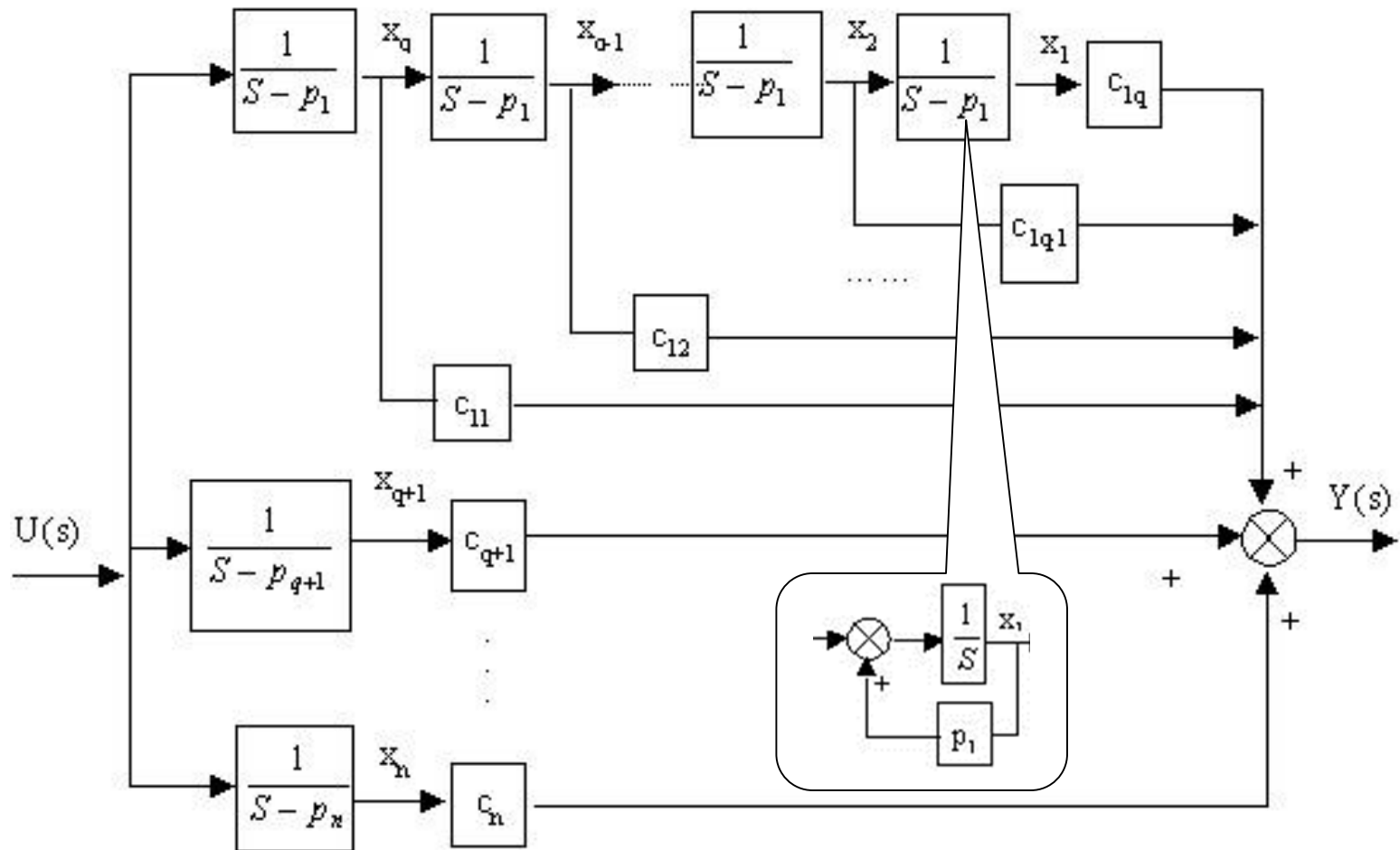
In which:

$$c_{1i} = \frac{1}{(q-i)!} \bullet \lim_{s \rightarrow p_1} \frac{d^{q-i}}{ds^{q-i}} [(s - p_1)^q G(s)] \quad i=1,2,\dots,q$$

$$c_j = \lim_{s \rightarrow p_j} [(s - p_j) G(s)] \quad j=q+1, q+2, \dots, n$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{c_{11}}{s - p_1} + \frac{c_{12}}{(s - p_1)^2} + \dots + \frac{c_{1q}}{(s - p_1)^q} + \frac{c_{q+1}}{s - p_{q+1}} + \dots + \frac{c_n}{s - p_n}$$

$$Y(s) = \frac{c_{11}}{s - p_1} U(s) + \frac{c_{12}}{(s - p_1)^2} U(s) + \dots + \frac{c_{1q}}{(s - p_1)^q} U(s) + \frac{c_{q+1}}{s - p_{q+1}} U(s) + \dots + \frac{c_n}{s - p_n} U(s)$$



Parallel Connection (Repeated roots)

$$Y(s) = \frac{c_{11}}{s - p_1} U(s) + \frac{c_{12}}{(s - p_1)^2} U(s) + \dots + \frac{c_{1q}}{(s - p_1)^q} U(s) + \frac{c_{q+1}}{s - p_{q+1}} U(s) + \dots + \frac{c_n}{s - p_n} U(s)$$

Select the state-variables as the output of the integral items in the diagram:

$$\begin{cases} \dot{x}_1 = p_1 x_1 + x_2 \\ \dot{x}_2 = p_1 x_2 + x_3 \\ \vdots \\ \dot{x}_{q-1} = p_1 x_{q-1} + x_q \\ \dot{x}_q = p_1 x_q + u \\ \vdots \\ \dot{x}_n = p_n x_n + u \end{cases}$$

$$y = c_{1q} x_1 + c_{1q-1} x_2 + \dots + c_{11} x_q + c_{q+1} x_{q+1} + \dots + c_n x_n$$

$$\left\{ \begin{array}{l} \dot{X} = \begin{bmatrix} p_1 & 1 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & p_1 & \cdots & \vdots & \vdots & \cdots & \vdots \\ \vdots & \vdots & \cdots & 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & p_1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & p_{q+1} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & p_n \end{bmatrix} X + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} U \\ Y = [c_{1q} \quad c_{1q-1} \quad \cdots \quad c_{11} \quad c_{q+1} \quad \cdots \quad c_n] X \end{array} \right.$$

System matrix A is Jordan Standard Form(约当标准型).

Ex.9-8 find the parallel connection of the follow system:

$$G(s) = \frac{4s^2 + 10s + 5}{s^3 + 5s^2 + 8s + 4}$$

Solution: the Denominator: $(s + 2)^2(s + 1)$

$$G(s) = \frac{4s^2 + 10s + 5}{(s + 2)^2(s + 1)} = \frac{c_{11}}{s + 2} + \frac{c_{12}}{(s + 2)^2} + \frac{c_3}{s + 1}$$

$$c_{12} = \lim_{s \rightarrow -2} (s + 2)^2 G(s) = -1$$

$$c_{11} = \frac{1}{(2 - 1)!} \lim_{s \rightarrow -2} \frac{d^{(2-1)}}{ds^{(2-1)}} [(s + 2)^2 G(s)] = \lim_{s \rightarrow -2} \frac{d}{ds} \left[\frac{4s^2 + 10s + 5}{(s + 1)} \right] = 5$$

$$c_3 = \lim_{s \rightarrow -1} (s + 1) G(s) = \lim_{s \rightarrow -1} \frac{4s^2 + 10s + 5}{(s + 2)^2} = -1$$

The state equation of the system by parallel connection:

$$\begin{cases} \dot{X} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} X + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u \\ y = [-1 \quad 5 \quad -1]X \end{cases}$$

2. State Space to Transfer Functions

State Space (Dynamic Equations) VS Transfer Functions

- State Space denote both the input/output relationship and the internal state variables of the system;
Transfer functions present the input/output relationship only.
- **From Transfer Functions to State Space:** system realization process, which is complicated and non-unique.
From State Space to Transfer Functions: simple and unique process.

❖ For SISO system: State-space to TF

The State-space representation of a SISO system:

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \\ \mathbf{y} = \mathbf{C}\mathbf{x} + Du \end{cases}$$

$\mathbf{x}, \dot{\mathbf{x}} \in R^{n \times 1}$; $\mathbf{A} \in R^{n \times n}$; $\mathbf{B} \in R^{n \times 1}$; $\mathbf{C} \in R^{1 \times n}$; D is a scalar quantity.

Assume the initial condition zero-input, and use Laplace Transform:

$$sX(s) = AX(s) + BU(s)$$

$$Y(s) = CX(s) + DU(s)$$

$$X(s) = (sI - A)^{-1}BU(s)$$

The Transfer Function is:

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

Ex.9-8 the state-space of the system is:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find its transfer function:

Solution: write the related matrices [A ,B ,C] at first:

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad C = [1 \quad 0]$$

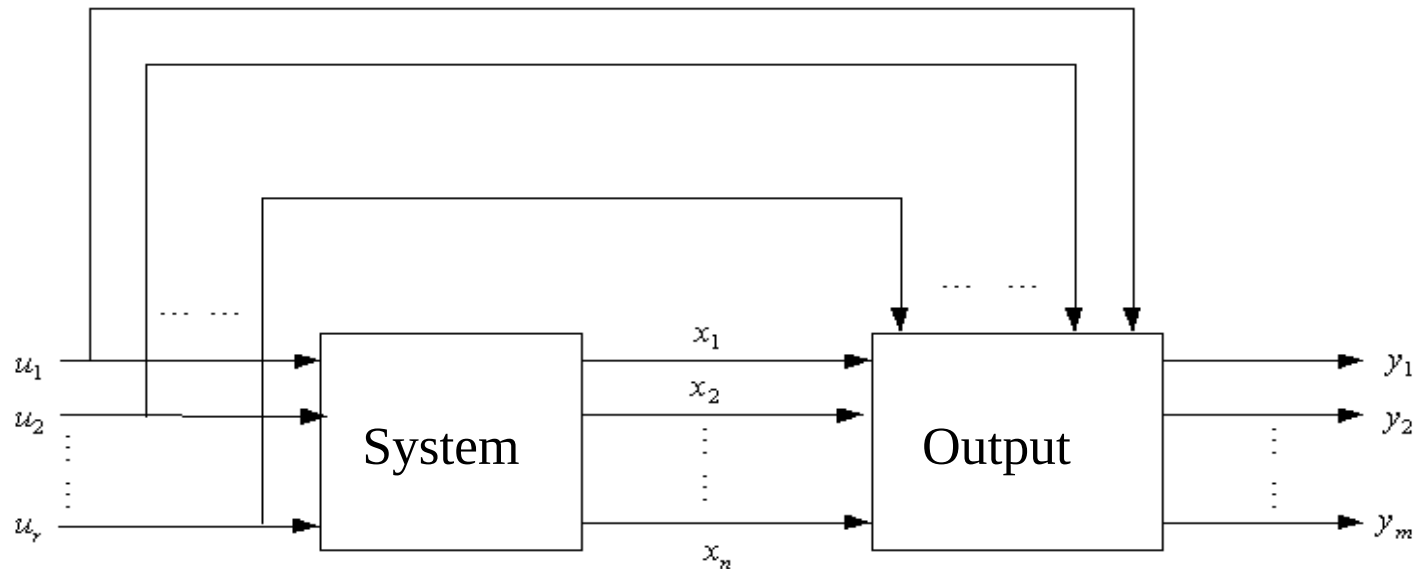
$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -1 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 1 & s+3 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s(s+3)+1} \begin{bmatrix} s+3 & 1 \\ -1 & s \end{bmatrix}$$

Transfer function:

$$\begin{aligned} G(s) &= \frac{Y(s)}{U(s)} = C(sI - A)^{-1} B \\ &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{s+3}{s(s+3)+1} & \frac{1}{s(s+3)+1} \\ \frac{-1}{s(s+3)+1} & \frac{s}{s(s+3)+1} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \frac{1}{s(s+3)+1} = \frac{1}{s^2 + 3s + 1} \end{aligned}$$

❖ For MIMO system: State-space to TF



$u_1, u_2, \dots, u_r$ ————— input signals of the system

$y_1, y_2, \dots, y_m$ ————— output signals of the system

$x_1, x_2, \dots, x_n$ ————— state variables of the system

The dynamic equations:
$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases}$$

Equations have the same formation with SISO system, however, the matrices B, C, D have different dimension.

$$\mathbf{x}, \dot{\mathbf{x}} \in \mathbf{R}^{n \times 1}, \quad \mathbf{U} \in \mathbf{R}^{r \times 1}, \quad \mathbf{Y} \in \mathbf{R}^{m \times 1},$$

$$\mathbf{A} \in \mathbf{R}^{n \times n}, \quad \mathbf{B} \in \mathbf{R}^{n \times r}, \quad \mathbf{C} \in \mathbf{R}^{m \times n}, \quad \mathbf{D} \in \mathbf{R}^{m \times r}$$

Laplace transformation:

$$\mathbf{G}(s) = \frac{\mathbf{Y}(s)}{\mathbf{U}(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$

$\mathbf{G} \in \mathbf{R}^{m \times r}$ — — — Transfer Function Matrix

Ex.9-9 The dynamic equations are:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find the Transfer function of the system.

Solution: $A = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}$ $C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $D = 0$

$$(sI - A)^{-1} = \begin{bmatrix} s & -1 \\ 0 & s+2 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix}$$

$$G(s) = C(sI - A)^{-1}B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix}$$

➤ the Eigen-equation and Eigenvalue of the system matrix A

The Eigen-equation of the system is:

$$|\lambda I - A| = 0$$

The Eigenvalue is one of its solutions.

Expand the equation: $|\lambda I - A| = 0$

we have the polynomial: $\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \dots + a_n = 0$

Then solve the equation we have its n eigenvalues.

e.g. $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$ then $|\lambda I - A| = \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{vmatrix}$

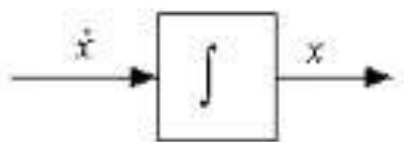
$$= \lambda^3 + 6\lambda^2 + 11\lambda + 6$$
$$= (\lambda + 1)(\lambda + 2)(\lambda + 3) = 0$$

3 eigenvalues are -1 、 -2 and -3 .

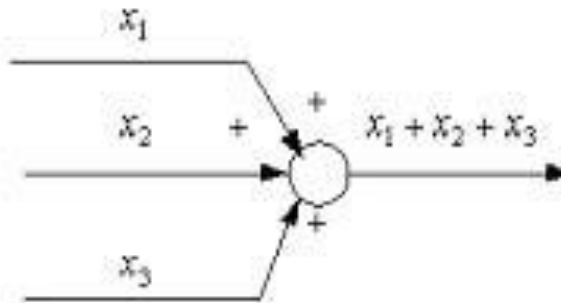
9.3.4 From State variable diagram of the System

- **State variable diagram:** the diagram description of the relationship of state variables, which is composed by the integral items, proportion items and sum symbols.

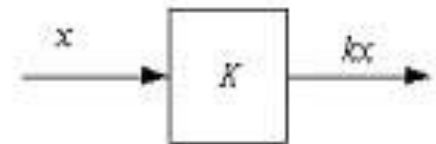
◆ The output of each integral item is one of the state variables of the system.



(a)



(b)



(c)

Ex.9-10 The Close-loop TF of the system is:

$$\frac{Y(s)}{U(s)} = \frac{s^2 + 3s + 2}{s(s^2 + 7s + 12)}$$

Draw the State variable diagram and find the state space representation of the system.

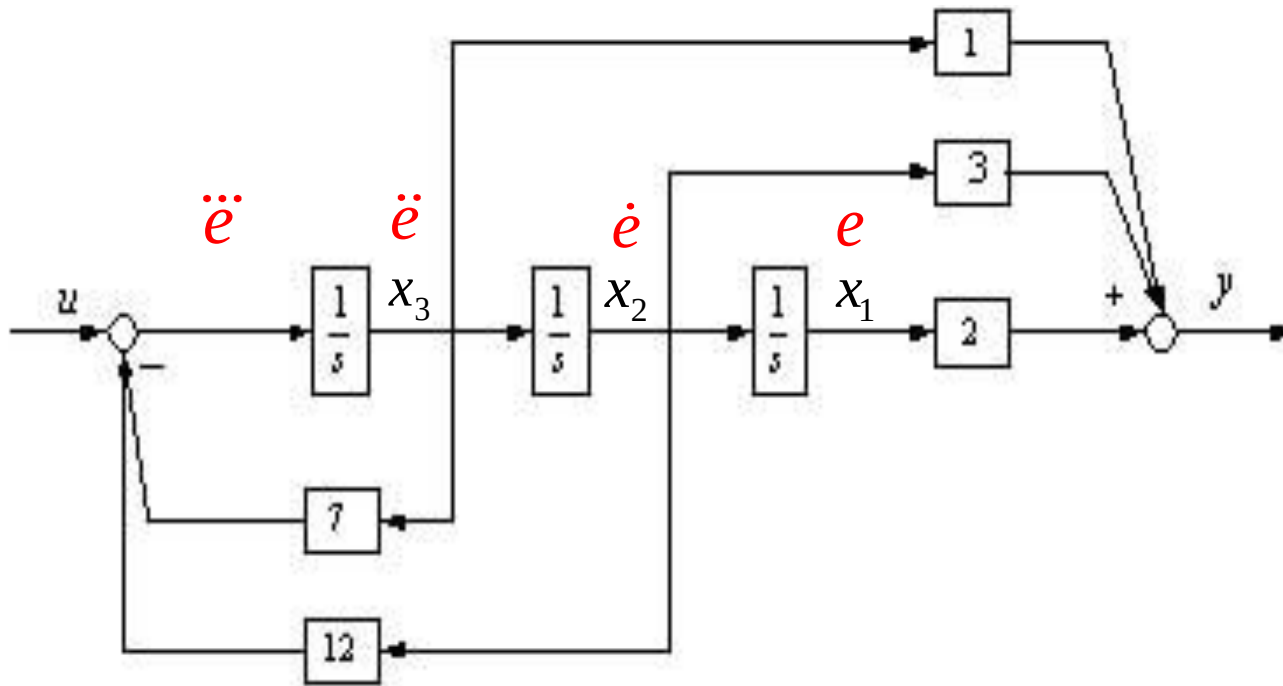
Solution: rewrite the close-loop TF of the system:

$$\frac{Y(s)}{U(s)} = \frac{s^2 + 3s + 2}{s^3 + 7s^2 + 12s}$$

Assume:

$$Y(s) = (s^2 + 3s + 2)E(s) \quad \Rightarrow \quad Y(s) = s^2 E(s) + 3sE(s) + 2E(s)$$
$$y(t) = \ddot{e}(t) + 3\dot{e}(t) + 2e(t)$$

$$U(s) = (s^3 + 7s^2 + 12s)E(s) \quad \Rightarrow \quad s^3 E(s) = U(s) - 7s^2 E(s) - 12sE(s)$$
$$\ddot{e}(t) = u(t) - 7\ddot{e}(t) - 12\dot{e}(t)$$



$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -12 & -7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \\ y = [2 \ 3 \ 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \end{cases}$$

9.3.5 Linear Transformation of State space

- State-space equation establishing method review:
Physics Mechanism / Differential Equations/ Transfer Functions / State-variable Diagram
- State-variables selection: **Non-unique**
- State-space equation: **Non-unique**
- The amount of the **independent state variables** in different state-space equation for a certain physics system: **Uniform**
- The connection of different state-space representations:
Linear Transformation

- Non-uniqueness of State-space Variables

Assume a State-space Equation:

$$\dot{x} = Ax + Bu$$

Linear Transform the state-variables: $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)$ into another group: $x_1, x_2, \dots, x_n$

We have

and also

$$x = P\bar{x}$$

$$\begin{cases} x_1 = p_{11}\bar{x}_1 + p_{12}\bar{x}_2 + \dots + p_{1n} \\ x_2 = p_{21}\bar{x}_1 + p_{22}\bar{x}_2 + \dots + p_{2n} \\ \dots\dots\dots \\ x_n = p_{n1}\bar{x}_1 + p_{n2}\bar{x}_2 + \dots + p_{nn} \end{cases}$$

$$P = \begin{bmatrix} p_{11} & p_{12} & \dots & p_{1n} \\ p_{21} & p_{22} & \dots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \dots & p_{nn} \end{bmatrix}$$

If P is a non-singular(非奇异) constant matrix: $|P| \neq 0$

The vector $\bar{x}$ is the state-variable vector of the system: $\dot{x} = Ax + Bu$

Proof:
$$\dot{\bar{x}} = P^{-1}AP\bar{x} + P^{-1}Bu$$

Symbolize: $A_1 = P^{-1}AP, B_1 = P^{-1}B$

then
$$\dot{\bar{x}} = A_1\bar{x} + B_1u$$

A_1 and A are similar matrices, the characteristic polynomial:

$$|sI - A_1| = |sI - A|$$

The state-equations from x and $\bar{x}$ have same eigenvalues

Result: for any control system, selection of state variables is non-unique. Any state variables from linear transformation, which satisfy the **non-singular condition** of the transform matrix P , are the appropriate state variables of the system.

Invariability(不变性) of linear transformation

$$\text{System: } \begin{cases} \dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \\ \mathbf{y} = C\mathbf{x} + D\mathbf{u} \end{cases} \quad \text{If } \mathbf{x} = P\bar{\mathbf{x}} \quad \begin{cases} \dot{\bar{\mathbf{x}}} = P^{-1}AP\bar{\mathbf{x}} + P^{-1}B\mathbf{u} \\ \mathbf{y} = CP\bar{\mathbf{x}} + D\mathbf{u} \end{cases}$$

(1) Invariability of eigenequations and eigenvalues

Eigenvalues after transformation:

Analysis:

$$\begin{aligned} |\lambda I - P^{-1}AP| &= |\lambda P^{-1}P - P^{-1}AP| = |P^{-1}\lambda P - P^{-1}AP| \\ &= |P^{-1}(\lambda I - A)P| = |P^{-1}| |\lambda I - A| |P| \\ &= |P^{-1}| |P| |\lambda I - A| = |P^{-1}P| |\lambda I - A| \\ &= |I| |\lambda I - A| = |\lambda I - A| \end{aligned}$$

Obviously, the eigenvalues of the system are same before and after the non-singular linear transformation.

(2) Invariant of system transfer function matrix

The transfer matrix after transformation:

$$\begin{aligned} G'(s) &= CP(sI - P^{-1}AP)^{-1}P^{-1}B + D \\ &= CP(P^{-1}sIP - P^{-1}AP)^{-1}P^{-1}B + D \\ &= CP[P^{-1}(sI - A)P]^{-1}P^{-1}B + D \\ &= CPP^{-1}(sI - A)^{-1}PP^{-1}B + D \\ &= C(sI - A)^{-1}B + D = G(s) \end{aligned}$$

The transfer matrix of the system is invariant before and after the non-singular linear transformation.

✓ Why need Linear Transformation?

Although there are infinite forms of state-space equations according to a certain system, which satisfy the non-singular linear transformation, only several kind of canonical forms are benefit for us:

- **Controllability Canonical Form**
- **Observability Canonical Form**
- **Diagonal Canonical Form**
- **Jordan Canonical Form.**

□ Controllability Canonical Form(能控标准型)

$$\left\{ \begin{array}{l} \dot{x} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u \\ y = [c_n \quad c_{n-1} \quad \cdots \quad c_1] x \end{array} \right.$$

Matrix A is also called companion matrix(友矩阵).

□ Observability Canonical Form(能观标准型)

$$\begin{cases} \dot{x} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ 0 & 1 & \cdots & 0 & -a_{n-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 1 & -a_1 \end{bmatrix} x + \begin{bmatrix} b_n \\ b_{n-1} \\ b_{n-2} \\ \vdots \\ b_1 \end{bmatrix} u \\ y = \begin{bmatrix} 0 & 0 & \cdots & 1 \end{bmatrix} x \end{cases}$$

□ Diagonal Canonical Form (I)

$$\left\{ \begin{array}{l} \dot{x} = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u \\ y = \begin{bmatrix} c_1 & c_2 & \cdots & c_n \end{bmatrix} x \end{array} \right.$$

□ Diagonal Canonical Form (II)

Select state variables: $X_i(s) = \frac{c_i}{s - \lambda_i} U(s)$

System Output is: $Y(s) = \sum_{i=1}^n X_i(s)$

Inverse Laplace Transformation:

$$\begin{cases} \dot{x} = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix} x + \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} u \\ y = [1 \quad 1 \quad \cdots \quad 1] x \end{cases}$$

□ Jordan Canonical Form

$$\left\{ \begin{array}{l} \dot{x} = \begin{bmatrix} \lambda_1 & 1 & 0 & & 0 \\ & \lambda_1 & 1 & & 0 \\ & & \lambda_1 & & 0 \\ \hline & & & \lambda_4 & \\ 0 & & & & \ddots \\ & & & & & \lambda_n \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u \\ y = [c_{11} \quad c_{12} \quad c_{13} \quad c_4 \quad \cdots \quad c_n] x \end{array} \right.$$

The Block of repeated poles $\begin{bmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 1 \\ 0 & 0 & \lambda_1 \end{bmatrix}$ is called **Jordan Block**.

(1) Outlines:

$$\left. \begin{array}{l} \dot{x} = Ax + Bu \\ y = Cx + Du \end{array} \right\} \xrightarrow[\text{Equivalent transforming}]{x = P\bar{x}} \left. \begin{array}{l} \dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u \\ \bar{y} = \bar{C}\bar{x} + \bar{D}u = y \end{array} \right\}$$

Non-canonical form Canonical form

(2) Relationship between the coefficient matrices

$$\because x = P\bar{x}, \quad \dot{x} = P\dot{\bar{x}},$$

P is N x N non-singular constant matrix

Substitute into the equations:

$$\left. \begin{array}{l} P\dot{\bar{x}} = AP\bar{x} + Bu \\ y = CP\bar{x} + Du \end{array} \right\} \Rightarrow \left. \begin{array}{l} \dot{\bar{x}} = P^{-1}AP\bar{x} + P^{-1}Bu \\ y = CP\bar{x} + Du \end{array} \right\}$$

$\Downarrow$

$$\bar{A} = P^{-1}AP, \quad \bar{B} = P^{-1}B, \quad \bar{C} = CP, \quad \bar{D} = D$$

- The constraint satisfied systems $\{A,B,C,D\}$ and $\{ \bar{A},\bar{B},\bar{C},\bar{D} \}$ are called **Similar Systems**;
- The related dynamic equations are called **Equivalent Dynamic Equations**;
- The linear transformation is called **Equivalent Transformation**.

The purpose of non-singular transformation is to transform the system matrix A into the canonical form $\bar{A}$

Some Common Linear transformation methods

(1) Transform A to Diagonal Form

(a) Assume a square matrix A with n different real eigenvalues:

$\lambda_1, \dots, \lambda_n$ which satisfy the follow eigen-equation.

$$\det(\lambda I - A) = |\lambda I - A| = 0$$



$$\bar{A} = P^{-1}AP = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$$

The non-singular transformation matrix $\mathbf{P}$ is composed by real eigenvectors: $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n]$

And the eigenvectors satisfy the equations:

$$A\mathbf{p}_i = \lambda_i\mathbf{p}_i \quad \text{or} \quad (\lambda_i I - A)\mathbf{p}_i = 0$$

(b) Assume a companion matrix(友矩阵) A with n different real eigenvalues: $\lambda_1, \dots, \lambda_n$

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}$$

Then the Vander-mode matrix(范德蒙特矩阵) P

$$P = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \cdots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \cdots & \lambda_n^2 \\ \vdots & \vdots & & \\ \lambda_1^{n-1} & & \cdots & \lambda_n^{n-1} \end{bmatrix}$$

can transform A to the diagonal matrix:

$$\Rightarrow \overline{A} = P^{-1}AP = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}$$

(c) Assume matrix A has m repeated eigenvalues: $\lambda_1 = \dots = \lambda_m$ and other $(n-m)$ different eigenvalues. If A still has m independent eigenvectors $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_m$ while solving the equation:

$$A\mathbf{p}_i = \lambda_i \mathbf{p}_i \quad (i = 1 \sim m)$$

the matrix A can be transformed to the diagonal form:

$$\Rightarrow \bar{A} = P^{-1}AP = \begin{bmatrix} \lambda_1 & & & & & 0 \\ & \ddots & & & & \\ & & \lambda_m & & & \\ & & & \lambda_{m+1} & & \\ & & & & \ddots & \\ 0 & & & & & \lambda_n \end{bmatrix}$$

$$P = [\mathbf{p}_1, \dots, \mathbf{p}_m \vdots \mathbf{p}_{m+1}, \dots, \mathbf{p}_n]$$

Ex.9-11: transform the follow state equation to the diagonal form

$$\dot{\mathbf{x}} = \begin{bmatrix} 2 & 4 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} u$$

Solution: characteristic equation

$$\begin{aligned} \det(\lambda I - A) &= |\lambda I - A| = \begin{vmatrix} \lambda - 2 & -4 & -5 \\ 0 & \lambda - 1 & 0 \\ 0 & 0 & \lambda - 1 \end{vmatrix} \\ &= (\lambda - 2)(\lambda - 1)^2 = 0 \end{aligned}$$

$$\lambda_1 = 2, \quad \lambda_2 = \lambda_3 = 1$$

$$(\lambda_1 I - A)\mathbf{p}_1 = 0 \quad \begin{bmatrix} 0 & -4 & -5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_{11} \\ p_{12} \\ p_{13} \end{bmatrix} = 0 \quad \mathbf{p}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$(\lambda_2 I - A)\mathbf{p}_2 = 0 \quad \begin{bmatrix} -1 & -4 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_{21} \\ p_{22} \\ p_{23} \end{bmatrix} = 0 \quad \mathbf{p}_2 = \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$$

$$(\lambda_2 I - A)\mathbf{p}_3 = 0 \quad \begin{bmatrix} -1 & -4 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_{31} \\ p_{32} \\ p_{33} \end{bmatrix} = 0 \quad \mathbf{p}_3 = \begin{bmatrix} 5 \\ 0 \\ -1 \end{bmatrix}$$

$$\therefore P = [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3] = \begin{bmatrix} 1 & 4 & 5 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\bar{A} = P^{-1}AP = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \bar{\mathbf{b}} = P^{-1}\mathbf{b} = \begin{bmatrix} 24 \\ -2 \\ -3 \end{bmatrix}$$

(2) transform A to Jordan form

(a) Assume matrix A has m repeated real eigenvalues:

$$\lambda_1 = \cdots = \lambda_m$$

others are $(n-m)$ different real eigenvalues.

Then solve the equation: $A\mathbf{p}_i = \lambda_i \mathbf{p}_i \quad (i = 1 \sim m)$

and receive only one independent eigenvector: $\mathbf{p}_1$

Matrix A can be transformed to Jordan Canonical Form only.

$$J = \bar{A} = P^{-1}AP = \begin{bmatrix} \begin{array}{cccc|ccc} \lambda_1 & 1 & & & & & \\ & \lambda_1 & \ddots & & & & \\ & & \ddots & 1 & & & \\ & & & \lambda_1 & & & \\ \hline & & & & \lambda_{m+1} & & \\ & & & & & \ddots & \\ & & & & & & \lambda_n \end{array} & \begin{array}{c} 0 \\ \\ \\ \\ 0 \\ \end{array} \end{bmatrix}$$

Jordan Block

$$P = [\mathbf{p}_1, \cdots, \mathbf{p}_m \vdots \mathbf{p}_{m+1}, \cdots, \mathbf{p}_n]$$

Here, $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_m$ are **generalized eigenvectors** (广义特征向量) and satisfy:

$$[\mathbf{p}_1, \cdots, \mathbf{p}_m] \begin{bmatrix} \lambda_1 & 1 & & \\ & \lambda_1 & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_1 \end{bmatrix} = A[\mathbf{p}_1, \cdots, \mathbf{p}_m]$$

$\mathbf{p}_{m+1}, \dots, \mathbf{p}_n$ are the eigenvectors corresponding to the (n-m) different eigenvalues.

(b) Assume **companion matrix A**, which is controllability canonical form matrix, and its m repeated eigenvalues:

$$\lambda_1 = \dots = \lambda_m$$

There is only one independent real eigenvector:

$$\mathbf{p}_1 = [1 \ \lambda_1 \ \lambda_1^2 \ \dots \ \lambda_1^{n-1}]^T$$

satisfy:

$$A\mathbf{p}_i = \lambda_i \mathbf{p}_i \quad (i = 1 \sim m)$$

which can transform A to Jordan form:

$$P = [\mathbf{p}_1 \ \frac{\partial \mathbf{p}_1}{\partial \lambda_1} \ \frac{\partial^2 \mathbf{p}_1}{\partial \lambda_1^2} \ \dots \ \frac{\partial^{m-1} \mathbf{p}_1}{\partial \lambda_1^{m-1}} \vdots \mathbf{p}_{m+1} \ \dots \ \mathbf{p}_n]$$

(c) Assume a matrix A has 5 repeated eigenvalues:

$$\lambda_1 = \dots = \lambda_5$$

Satisfy: $A\mathbf{p}_i = \lambda_i \mathbf{p}_i \quad (i = 1 \sim 5)$

With 2 independent real eigenvectors: $\mathbf{p}_1$ and $\mathbf{p}_2$.

Other $(n-5)$ eigenvalues are different.

The matrix A can be transformed to the Jordan form:

$$J = \bar{A} = P^{-1}AP = \begin{bmatrix} \lambda_1 & 1 & & & & & \\ & \lambda_1 & 1 & & & & \\ & & \lambda_1 & & & & \\ \hline & & & \lambda_1 & 1 & & \\ & & & & \lambda_1 & & \\ & & & & & \lambda_{m+1} & \\ & & & & & & \ddots \\ & & & & & & & \lambda_n \end{bmatrix}$$

There are 2 upper Jordan blocks in J , in which:

$$P = \begin{bmatrix} p_1 & \frac{\partial p_1}{\partial \lambda_1} & \frac{\partial^2 p_1}{\partial \lambda_1^2} & \vdots & p_2 & \frac{\partial p_2}{\partial \lambda_1} & \vdots & p_6 & \cdots & p_n \end{bmatrix}$$

(3) Transform the controllable system to Controllability Canonical Form

The controllability canonical form of a single input linear time-invariant system state equations:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u$$

- The corresponding **controllability matrix S** is a Right Lower Triangular matrix with the main diagonal elements 1.
- Thus $\det S \neq 0$, the system is controllable, and A, b are called **controllability canonical form**.
- The controllability matrix S is:

$$S = [b \quad Ab \quad \dots \quad A^{n-1}b] = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & 0 & \dots & 1 & -a_{n-1} \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 1 & \dots & \times & \times \\ 0 & 1 & -a_{n-1} & \dots & \times & \times \\ 1 & -a_{n-1} & -a_{n-2} & \dots & \times & \times \end{bmatrix}$$

- Any controllable system, if its A , b are not controllability canonical form, they can be transformed to the canonical form by appropriate transforming method.
- Assume a dynamic system: $\dot{x} = Ax + bu$
- Execute the P^{-1} transformation:

$$x = P^{-1}z$$

and we have

$$\dot{z} = PAP^{-1}z + Pbu$$

Satisfy:

$$PAP^{-1} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}, \quad Pb = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

Analyze the transformation matrix: P

Assume P is

$$P = \begin{bmatrix} p_1^T & p_2^T & \cdots & p_n^T \end{bmatrix}^T$$

Based on the matrix A, P should satisfy:

$$\begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_{n-1} \\ p_n \end{bmatrix} A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_{n-1} \\ p_n \end{bmatrix}$$

$$\left\{ \begin{array}{l} p_1 A = p_2 \\ p_2 A = p_3 \\ \vdots \\ p_{n-1} A = p_n \\ p_n A = -a_0 p_1 - a_1 p_2 - \cdots - a_{n-1} p_n \end{array} \right.$$

Then:

$$\left\{ \begin{array}{l} p_1 A = p_2 \\ p_2 A = p_1 A^2 = p_3 \\ \vdots \\ p_{n-1} A = p_1 A^{n-1} = p_n \end{array} \right.$$

Transformation matrix:

$$P = \begin{bmatrix} p_1 \\ p_1 A \\ \vdots \\ p_1 A^{n-1} \end{bmatrix}$$

Then based on vector b , we have:

$$Pb = \begin{bmatrix} p_1 \\ p_1 A \\ \vdots \\ p_1 A^{n-1} \end{bmatrix} b = \begin{bmatrix} p_1 b \\ p_1 Ab \\ \vdots \\ p_1 A^{n-1} b \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$p_1 [b \quad Ab \quad \dots \quad A^{n-1} b] = [0 \quad \dots \quad 0 \quad 1]$$

Thus $p_1 = [0 \quad \cdots \quad 0 \quad 1] [b \quad Ab \quad \cdots \quad A^{n-1}b]^{-1}$

It seems that p_1 is the last row of the inverse controllability matrix

Therefore, the solution of transformation matrix P^{-1} :

(i) Find the controllability matrix $S = [b \quad Ab \quad \cdots \quad A^{n-1}b]$

(ii) Find the inverse matrix S^{-1} , which is:

$$S^{-1} = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1n} \\ S_{21} & S_{22} & \cdots & S_{2n} \\ \vdots & \vdots & & \vdots \\ S_{n1} & S_{n2} & \cdots & S_{nn} \end{bmatrix}$$

(iii) Take out the last row of S^{-1} , (the n^{th} row) and compose the vector:

$$p_1 = [S_{n1} \quad S_{n2} \quad \cdots \quad S_{nn}]$$

(iv) Construct matrix P

$$P = \begin{bmatrix} p_1 \\ p_1 A \\ \vdots \\ p_1 A^{n-1} \end{bmatrix}$$

(v) Then, P^{-1} is required transforming matrix from non-canonical form to controllability canonical form.

(4) Canonical Form of SISO system– From TF

The state space description of the dynamic system:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases} \quad x = P\bar{x}$$

- ✓ Consequently, according to transformation matrix P , we can transform the above system to the canonical forms we need.
- ✓ Here, we consider about the transformation using the transfer functions.

The required transfer function is:

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

The canonical state space forms of **controllability form**, **observability form** and **diagonal form** (or **Jordan form**) are given as follow.

(1) Controllability Canonical Forms

$$\left\{ \begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} &= \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u \\ \\ y &= [b_n - a_n b_o \quad b_{n-1} - a_{n-1} b_o \quad \cdots \quad b_1 - a_1 b_o] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u \end{aligned} \right.$$

(2) Observability Canonical Forms

$$\left\{ \begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} &= \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ 0 & 1 & \cdots & 0 & -a_{n-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 1 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} b_n - a_n b_0 \\ b_{n-1} - a_{n-1} b_0 \\ b_{n-2} - a_{n-2} b_0 \\ \vdots \\ b_1 - a_1 b_0 \end{bmatrix} u \\ \\ y &= [0 \ 0 \ \cdots \ 0 \ 1] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + b_o u \end{aligned} \right.$$

(3) Diagonal Canonical Form

If the roots of denominator polynomial are different, transfer function can be written as follow:

$$\frac{Y(s)}{U(s)} = \frac{b_o s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{(s + p_1)(s + p_2) \dots (s + p_n)} = b_o + \frac{c_1}{s + p_1} + \frac{c_2}{s + p_2} + \dots + \frac{c_n}{s + p_n}$$

The canonical form is:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -p_1 & & & 0 \\ & -p_2 & & \\ & & \ddots & \\ 0 & & & -p_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u$$

$$y = [c_1 \ c_2 \ \dots \ c_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_o u$$

(4) Jordan Canonical Form

If there are **repeated eigenvalues** in denominator, transform the diagonal form to Jordan form. Assume 3 repeated roots are $-p_1 = -p_2 = -p_3$ and others are different.

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{(s + p_1)^3 (s + p_4)(s + p_5) \dots (s + p_n)}$$

$$= b_0 + \frac{c_1}{(s + p_1)^3} + \frac{c_2}{(s + p_1)^2} + \frac{c_3}{(s + p_1)} + \frac{c_4}{s + p_4} + \dots + \frac{c_n}{s + p_n}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -p_1 & 1 & 0 & 0 & \dots & 0 \\ 0 & -p_1 & 1 & \vdots & & \vdots \\ 0 & 0 & -p_1 & 0 & & 0 \\ 0 & \dots & 0 & -p_4 & & 0 \\ \vdots & & \vdots & & \ddots & \\ 0 & \dots & 0 & 0 & & -p_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u$$

$$y = [c_1 \ c_2 \ \dots \ c_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_o u$$

Ex.9-12 consider the follow transfer function of a certain system:

$$\frac{Y(s)}{U(s)} = \frac{s + 3}{s^2 + 3s + 2}$$

try to find its **controllability canonical form**, **observability canonical form**, and **diagonal form**.

Solution:

Its controllability canonical form is:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$
$$y(t) = [3 \quad 1] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Observability canonical form is:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} u(t)$$
$$y(t) = [0 \quad 1] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Diagonal canonical form is:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t)$$
$$y(t) = [2 \quad -1] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$